

THE POWER OF HUMAN NETWORKING

Why are we here?



IDEA EXCHANGE

Professional networking is the “pre-orchestration” layer for any successful technology implementation.



THE CATALYST

Personal connections lead to “Agentic” outcomes — where the whole is greater than the sum of its parts.

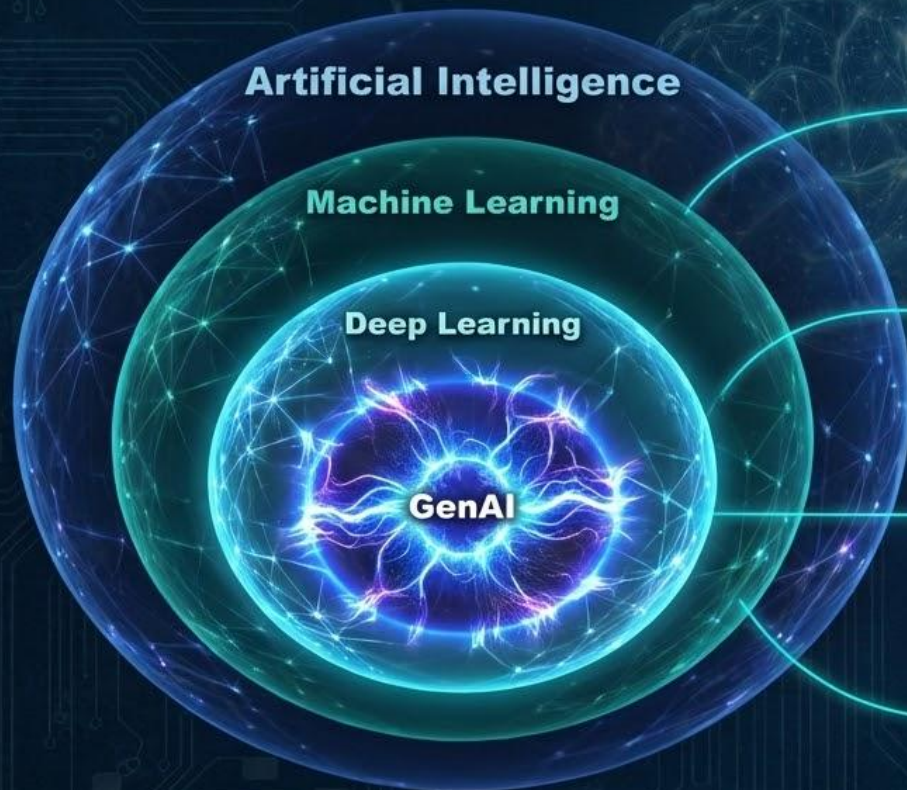


INTERACTIVE MOMENT

“Look to your left and right. That’s your human ‘multi-agent system’ for today.”

DEMYSTIFYING AI

The 'Brain' Hierarchy



Artificial Intelligence

The broad goal of machines mimicking human like intelligence

Machine Learning

Statistical learning from data (e.g., traditional claims processing)

Deep Learning

Complex architectures mimicking neural pathways like a human memory

Generative AI

Models that create new artifacts (text, code, schemas)

QUICK STATUS CHECK

Let's See Where Everyone Stands

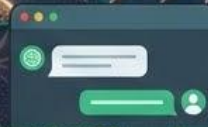


RAISE YOUR HAND IF...

BEGINNER



You've used ChatGPT, Claude, or Gemini at least once?



INTERMEDIATE



You've heard of "Prompt Engineering" or written custom prompts?



PRACTITIONER



You know what "Tokens" or "Context Window" means?



ADVANCED



You've heard of "AI Agents" or "Agentic AI"?



EXPERT



You've built something with LangChain, MCP, or an Agent framework?



Don't worry — we'll cover ALL these concepts in the next few slides!

THE LANGUAGE OF AI

How Models Read and Think

"Hello, how are you today?"



TOKENIZATION

Hello

,

how

are

you

today

?

7 tokens = ~\$0.00002



TOKENS

The basic units AI reads — like syllables.
1 token $\approx$ 4 characters or $\frac{3}{4}$ of a word.
You pay per token!



CONTEXT WINDOW

The AI's "memory" for a conversation.
GPT-4: 128K tokens.
Claude: 200K tokens. Bigger = more context!



SYSTEM INSTRUCTIONS

Hidden rules you give the AI before it responds. "You are a helpful assistant that speaks formally."

CONTEXT WINDOW VISUALIZATION



System Instructions (hidden): Hidden rules you give the AI before it responds. 'You are a helpful assistant that speaks formally.'



User: "Help me write an email..."



AI: "Sure! Here's a draft..."



KEY INSIGHT: Every conversation starts fresh — the AI has **NO** memory beyond the context window

EMBEDDINGS & VECTORS

How AI Understands Meaning

THE WORD-TO-NUMBER MAGIC

"King"



[0.23, -0.87, 0.45...]

SEMANTIC SIMILARITY

"King" ↔ "Queen"

0.92 similar

"King" ↔ "Monarch"

0.89 similar

"King" ↔ "Apple"

0.12 similar



EMBEDDINGS

Converting words/sentences into numbers (vectors) that capture MEANING. Similar concepts = similar numbers!



VECTORIZATION

The process of converting ANY data (text, images, audio) into numerical vectors for AI processing.



VECTOR DATABASE

A special database (Pinecone, Weaviate, Chroma) that stores embeddings and finds similar items FAST. Powers RAG!



KEY INSIGHT: Embeddings let AI understand that "automobile" and "car" mean the same thing!

TRANSFORMERS

The Engine Behind Modern AI

THE ATTENTION MECHANISM

"The cat sat on the mat because it was tired"



✓ Transformer knows "it" = "cat"!



TRANSFORMERS

The neural network architecture (2017) that powers GPT, Claude, Gemini. Uses "attention" to understand relationships between ALL words at once!



SELF-ATTENTION

Lets the model look at ALL parts of input simultaneously and decide which parts are most relevant to each other.



LLM (Large Language Model)

A Transformer trained on massive text data (books, web, code). "Large" = billions of parameters. GPT-4 has ~1.7 trillion!



KEY INSIGHT: Transformers revolutionized AI by processing entire sequences in parallel, not one word at a time

WORKING WITH AI

Prompt Engineering & Fine-Tuning

PROMPT ENGINEERING:

The art of crafting inputs to get better outputs

✗ BAD PROMPT

"Write about dogs"



✓ GOOD PROMPT

"Write a 200-word blog post for pet owners about the top 3 health benefits of daily dog walking. Use a friendly tone and include one statistic."

- 📋 Be specific with format & length
- 👤 Define the role/persona
- 📄 Provide examples (few-shot)
- 🔗 Chain of thought reasoning



FINE-TUNING

Training an existing model on YOUR data to specialize it. Expensive but powerful. Example: Fine-tune GPT on your company's legal documents.



RAG (Retrieval-Augmented Generation)

Instead of fine-tuning, FETCH relevant docs at query time and include them in context. Cheaper & always up-to-date!



AGENTS

AI that can TAKE ACTIONS: browse web, call APIs, run code, use tools. Not just chat — actually DO things!



KEY INSIGHT: Most enterprise use cases need RAG + Agents, not fine-tuning!

AI TERMINOLOGY CHEAT SHEET

Your Quick Reference Guide

Token

Basic unit AI reads (~4 chars)

Context Window

AI's memory limit per conversation

System Prompt

Hidden instructions for AI behavior

Embedding

Words → numbers capturing meaning

Vector DB

Database for similarity search

Transformer

The architecture behind GPT/Claude

LLM

Large Language Model (billions of params)

Prompt Engineering

Crafting better inputs for AI

Fine-Tuning

Training model on custom data

RAG

Fetch docs → include in context

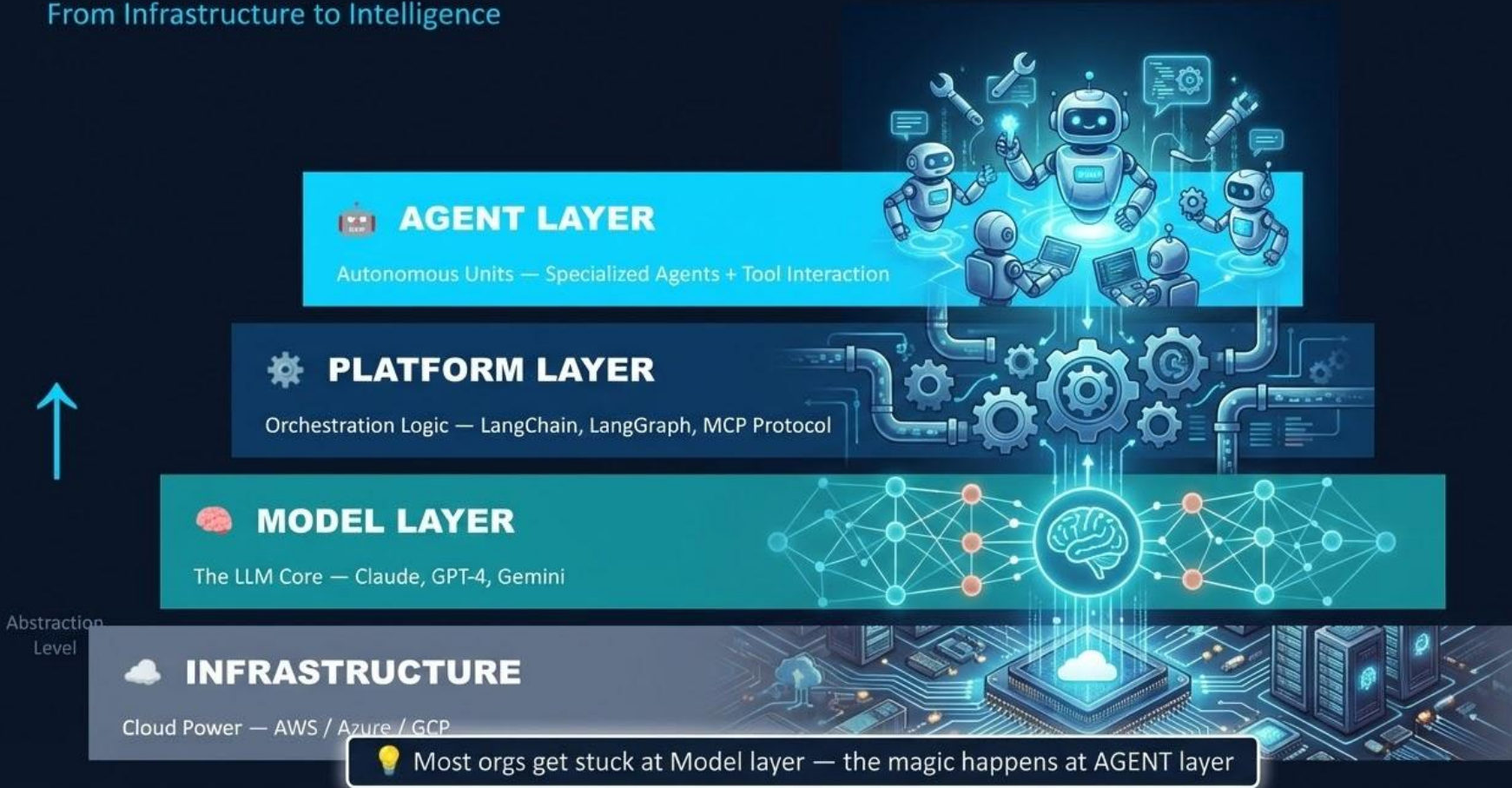
Agent

AI that takes actions, uses tools

✓ **NOW YOU'RE READY FOR THE DEEP DIVE INTO AGENTIC AI!**

THE 4-LAYER AGENTIC STACK

From Infrastructure to Intelligence



AI AGENT vs AGENTIC AI

The OIPA Reasoning Loop

AI AGENT

Reactive Tool



Follows commands



Single task focus



No memory across tasks



Limited reasoning



VS

AGENTIC AI

Proactive System



Uses reasoning loop



Multi-step planning



Shared context memory



Autonomous execution



THE OIPA REASONING LOOP



OBSERVE

Perceive environment
(e.g., Jira ticket)



INTERPRET

Understand user
intent & context



PLAN

Strategize sequence
of steps



ACT

Execute using
external tools

REASONING LOOP

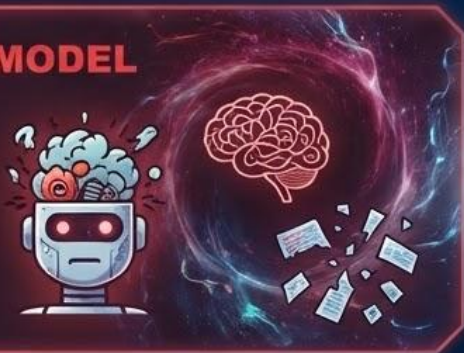


THE PHILOSOPHY OF SPECIALIZATION

“Hiring one generalist is chaos;
hiring a team of specialists is professional.”

✗ THE CHAOS MODEL

- Single massive prompt
- One AI for everything
- Context overload
- Unreliable outputs



VS

✓ THE SPECIALIST MODEL

- Specialized agents
- Domain expertise
- Focused context
- Reliable handoffs



👛 REAL-WORLD PARALLEL

You don't ask a developer to do legal compliance — you use a Specialized Agent. Agentic AI mirrors how professional teams actually operate.



ENTERPRISE CHALLENGES IN COORDINATION

⚡ **KEY INSIGHT:** Autonomy is easy; **COORDINATION** is the challenge.

RATE LIMITS



⚙️ Managing API traffic across an organization

🔧 **SOLUTION:** Token budgeting, queue management, fallback models

CONTEXT LIMITS



🔗 Preventing "tribal knowledge" loss across pipeline

📄 **SOLUTION:** Vector DBs, shared memory, artifact repositories

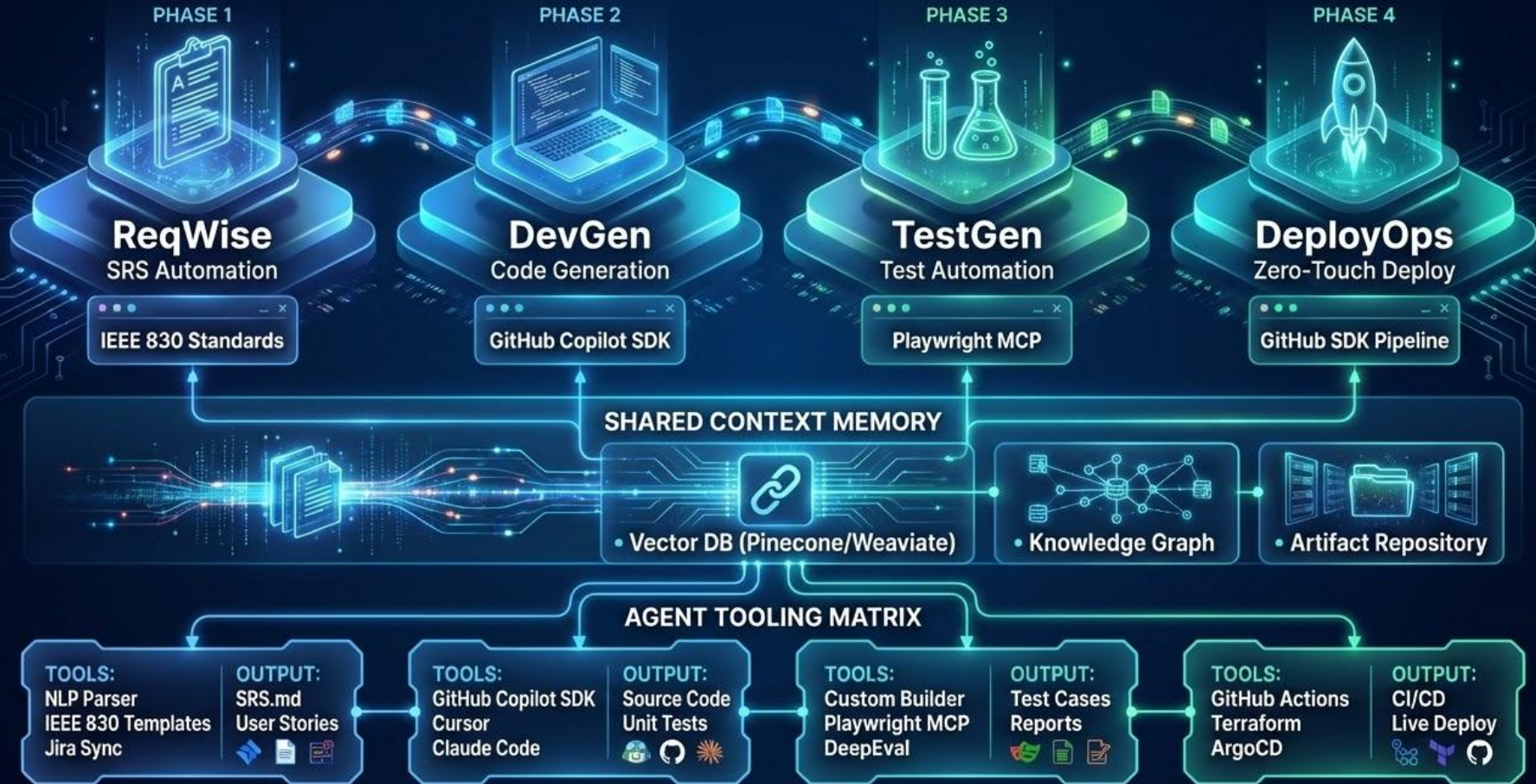
REALITY CHECK



🎯 Strategies for "Grounding" to stop hallucinations

📋 **SOLUTION:** Validation gates, human-in-the-loop, source verification

AgentOps.AI THE MASTER SDLC MODEL



THE COMPLIANCE BOT

Beyond the Code





LIVE DEMO

THE AUTONOMOUS FACTORY IN ACTION

THE BIG QUESTION WILL AI REPLACE MY JOB?

THE REALITY



We still need
Human-in-the-Loop to
avoid hallucinations and
protect business
reputation.

MARKET LESSON



Organizations are moving
BACK to human tele-callers
after voice agent failures.
Reputation matters more
than automation speed.

THE VERDICT



"AI won't replace you, but
someone USING AI will.
Evolve your networking
skills to include AI as your
partner."

CONCLUSION

KEY MESSAGE

Focus on BUSINESS OUTCOMES, not technology hype.
Agentic AI is a means to an end of operational excellence.



Specialists > Generalists — in AI and in life



Coordination is the real challenge, not autonomy



Human-in-the-Loop is essential, not optional



Evolve WITH AI — don't compete against it

"Let's build the autonomous factory of tomorrow, today."